

Project: Exam Result Analysis Dashboard

Goal: Visualize how students performed in different subjects, analyze pass/fail rates, compare gender/class-wise performance, and identify top-performing students.

Sample Dataset (You can create this in Excel or CSV)

Student_ID	Name	Class	Gender	Subject	Marks	Max_Marks
S001	Aman	10	Male	Math	85	100
S002	Priya	10	Female	Science	78	100
S003	Ravi	10	Male	English	92	100

Create your file like this and **save it as exam_results.csv** or Excel.

Step-by-Step Guide

Step 1: Open Power BI Desktop

- ❖ Open Power BI on your PC.
- ❖ You will see a blank canvas with options on top.

Step 2: Import the Dataset

- ❖ Click “**Get Data**” → Select “**Text/CSV**” or “**Excel**”.
- ❖ Browse and choose your file exam_results.csv or .xlsx.
- ❖ A preview opens. Click “**Load**”.

Step 3: Open Power Query Editor (Optional)

- ❖ Click “**Transform Data**” to open the **Power Query Editor**.
- ❖ Things to check:
 - ❖ **Marks and Max_Marks** → Should be in **Whole Number** format.
 - ❖ **Class** → Whole Number or Text.
 - ❖ **Gender, Subject** → Text format.

If everything looks good, click “**Close & Apply**” to bring the data into Power BI.

Step 4: Add Calculated Columns (Very Easy)

Click on “**Modeling**” → **New Column**

4.1 Percentage Column

Percentage = ('exam_results'[Marks] / 'exam_results'[Max_Marks]) * 100

4.2 Result Column

Result = IF('exam_results'[Percentage] >= 40, "Pass", "Fail")

These new columns will now be visible in your table.

Step 5: Add DAX Measures (Optional but Powerful)

Go to **Modeling** → **New Measure**

5.1 Total Students

Total Students = DISTINCTCOUNT('exam_results'[Student_ID])

5.2 Average Marks

Average Marks = AVERAGE('exam_results'[Marks])

5.3 Pass Count

Pass Count = CALCULATE(COUNTROWS('exam_results'), 'exam_results'[Result] = "Pass")

5.4 Fail Count

Fail Count = CALCULATE(COUNTROWS('exam_results'), 'exam_results'[Result] = "Fail")

Step 6: Create Visuals – One by One

6.1 KPI Cards (Big Numbers)

❖ Go to **Visualizations** → **Card**

❖ Drag the following fields to make 4 cards:

- ❖ Total Students
- ❖ Average Marks
- ❖ Pass Count
- ❖ Fail Count

6.2 Column Chart – Average Marks by Subject

- ❖ Use **Clustered Column Chart**
- ❖ Axis: Subject
- ❖ Value: Marks → Change to "Average"

6.3 Donut Chart – Pass vs Fail

- ❖ Use **Donut Chart**
- ❖ Legend: Result
- ❖ Values: Count of Student_ID

6.4 Bar Chart – Gender-wise Performance

- ❖ Axis: Gender
- ❖ Value: Average of Marks

6.5 Line Chart – Class-wise Average

- Axis: Class
- Value: Average of Marks

6.6 Table View – Detailed Student List

- ❖ Use **Table visual**
- ❖ Add:
 - ❖ Student_ID
 - ❖ Name
 - ❖ Class
 - ❖ Subject
 - ❖ Marks

- ❖ Percentage
- ❖ Result

Step 7: Add Filters (Slicers)

Go to **Visualizations** → **Slicer** and add:

- ❖ Class
- ❖ Gender
- ❖ Subject
- ❖ Result (Pass/Fail)

This allows you to click and filter the whole dashboard.

Step 8: Design and Format

- ❖ Add a **Text Box** → Type: “Exam Result Analysis Dashboard”
- ❖ Use **Themes**: Go to View → Themes → Pick a color scheme
- ❖ Align visuals → View → Turn on "Snap to Grid"
- ❖ Format numbers:
 - ✓ Percentage → No decimals
 - ✓ Marks → Whole numbers

Step 9: Save Your Dashboard

- ❖ Go to **File** → **Save As**
- ❖ Name it: Exam_Result_Analysis.pbix

Step 10: Export or Publish

You can:

- ❖ **Export to PDF**: File → Export → PDF
- ❖ **Export to PowerPoint**: File → Export → PowerPoint
- ❖ **Publish Online**: Home → Publish → Power BI Service (need sign in)

Final Dashboard Components

Feature	Visual Type
Total Students	Card
Average Marks	Card
Pass / Fail Count	Card
Avg Marks per Subject	Column Chart
Result Distribution	Donut Chart
Gender-wise Performance	Bar Chart
Class-wise Trend	Line Chart
Student Table	Table
Filters	Slicers (Class, Subject, Gender, Result)

Learned:

- ❖ **How to import and clean student data**
- ❖ **Create percentage and pass/fail columns**
- ❖ **Create KPIs and interactive visuals**
- ❖ **Add filters (slicers) to analyze by subject, gender, class**
- ❖ **Save and export your dashboard**